

Design of a Lead-Lag Compensator for the TRMS Pitch Channel Using the Root Locus Technique

CONTROL SYSTEMS

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1. Introduction

The Twin Rotor Multi-Input Multi-Output System (TRMS) is a laboratory model widely used to study the dynamics and control of nonlinear aerospace systems. Due to the coupling between its pitch and yaw motions, the TRMS exhibits complex dynamic behavior, making it an ideal platform for evaluating classical control techniques.

In this project, the TRMS pitch dynamics are analyzed by converting the mathematical state-space model into a transfer function. The open-loop system is examined through pole-zero analysis and the root locus method to determine its stability characteristics. Based on the design requirements, a lead-lag controller is developed to stabilize the system and improve its transient response.

The controller is designed using the root locus technique and verified through MATLAB simulations. The primary objective is to obtain a stable closed-loop system with low overshoot while evaluating the achievable settling time. The simulation results are analyzed to assess the effectiveness of the designed controller and to discuss the trade-offs between stability, overshoot, and response speed.

2. Objectives

The objectives of this project are:

- To obtain the transfer function of the TRMS pitch system from the given state-space model.
- To analyze the open-loop system by determining its poles, zeros, stability, and root locus characteristics.
- To identify the breakaway point, imaginary-axis crossing, and stable gain range using the root locus method.
- To establish the desired transient response specifications in terms of overshoot and settling time.
- To design a lead-lag controller using the root locus technique to improve the system response.
- To implement and verify the controller using MATLAB simulations.
- To evaluate the closed-loop performance by analyzing stability, overshoot, settling time, and pole locations.
- To investigate the trade-off between overshoot reduction and settling time for the designed lead-lag controller.

3. Design Methodology

The design of the TRMS pitch controller was carried out using a systematic classical control approach based on the Root Locus technique. The controller was first designed analytically through mathematical calculations and then verified using MATLAB simulations. This methodology provides a clear understanding of the system dynamics before validating the design through simulation.

The controller design procedure is summarized as follows.

Step 1: State-Space to Transfer Function

The mathematical model of the TRMS was provided in state-space form. Since the Root Locus technique requires a transfer function, the state-space model was converted using the standard state-space transformation.

$$G(s) = C(sI - A)^{-1}B + D$$

The resulting transfer function was then used for all controller design calculations.

Step 2: Open-Loop Analysis

The transfer function was analyzed to determine the open-loop poles, zeros, and stability characteristics of the system.

The characteristic equation of the closed-loop system is

$$1 + KG(s) = 0$$

Using this equation, the centroid, asymptote angles, breakaway point, imaginary-axis crossing, and stable gain range were determined through Root Locus analysis.

Step 3: Design Specifications

The desired transient response specifications were defined in terms of maximum overshoot and settling time.

The damping ratio was calculated using

$$\zeta = -\ln(M_p) / \sqrt{[\pi^2 + (\ln(M_p))^2]}$$

where

M_p = Maximum Overshoot

The required natural frequency was then determined using

$$\omega_n = 4 / (\zeta T_s)$$

T_s = Settling Time

These equations establish the desired dynamic characteristics of the compensated system.

Step 4: Desired Pole Location

The desired dominant closed-loop poles were determined using

$$s = -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}$$

These poles represent the desired operating point for the compensated system.

Step 5: Lead Compensator Design

If the desired poles do not lie on the uncompensated Root Locus, a Lead Compensator is introduced to provide the required phase compensation.

The Lead Compensator is represented by

$$G_{\text{Lead}}(s) = (s + z_1) / (s + p_1)$$

where

$$p_1 > z_1$$

The controller parameters are selected using the Root Locus Angle Criterion.

Step 6: Gain Calculation

After selecting the Lead Compensator, the controller gain is determined using the Root Locus Magnitude Criterion.

$$|KG(s)| = 1$$

The calculated gain ensures that the desired dominant poles satisfy the Root Locus conditions.

Step 7: Lag Compensator Design

To improve the steady-state response, a Lag Compensator is introduced.

The Lag Compensator is represented by

$$G_{\text{Lag}}(s) = (s + z_2) / (s + p_2)$$

where

$$z_2 > p_2$$

The lag network increases the low-frequency gain while producing minimal effect on the transient response.

Step 8: Complete Lead-Lag Controller

The overall controller is obtained by combining the gain, Lead Compensator, and Lag Compensator.

$$G_c(s) = K \times G_{\text{Lead}}(s) \times G_{\text{Lag}}(s)$$

$$G_c(s) = K \times [(s + z_1)(s + z_2)] / [(s + p_1)(s + p_2)]$$

Step 9: MATLAB Verification

After completing the mathematical design, MATLAB was used to verify the controller performance. The compensated system was evaluated using Root Locus analysis, step response, pole-zero maps, and Bode plots. The simulation results were then compared with the desired design specifications to evaluate the effectiveness of the proposed Lead-Lag controller.

4. System Modeling and Controller Design Approach

4.1 Mathematical Model

The Twin Rotor Multi-Input Multi-Output System (TRMS) is a nonlinear laboratory system that emulates the dynamic behavior of a helicopter. The system consists of two rotors that independently control the pitch and yaw motions. Due to the interaction between these two axes, the TRMS exhibits coupled dynamics and is widely used for the analysis and design of control systems.

The mathematical model of the TRMS was provided in state-space form. Since the Root Locus technique is based on transfer function analysis, the state-space representation was converted into an equivalent transfer function representing the pitch dynamics of the TRMS.

The resulting transfer function is

$$G(s) = 1.359 / (s^3 + 0.9973s^2 + 4.786s + 4.278)$$

The obtained transfer function is a third-order system with no finite zeros and three open-loop poles. This transfer function forms the mathematical model used throughout the controller design process.

4.2 Open-Loop Analysis

The open-loop transfer function was analyzed to determine the inherent characteristics of the TRMS pitch system before controller design. The poles, zeros, system order, and stability were evaluated to understand the system dynamics.

Results

Parameter	Value
System Order	3
Open-Loop Poles	$-0.0441 \pm j2.1688$, -0.9091
Open-Loop Zeros	None
Open-Loop Stability	Stable

The analysis shows that the plant is open-loop stable but possesses lightly damped complex poles, indicating oscillatory behavior that requires compensation for improved transient performance.

4.3 Root Locus Analysis

The Root Locus method was employed to examine the movement of the closed-loop poles as the controller gain varied. The root locus characteristics were determined analytically to establish the stability limits of the uncompensated system.

Results

Parameter	Value
Number of Branches	3
Number of Asymptotes	3
Centroid	-0.3324
Asymptote Angles	60° , 180° , 300°
Breakaway Point	No Real Breakaway Point
Imaginary Axis Crossing	$\pm j2.1877$

Critical Gain	0.3643
Stable Gain Range	$0 < K < 0.3643$

The root locus analysis indicated that the desired closed-loop pole locations could not be achieved using gain adjustment alone. Consequently, a compensator was required to reshape the root locus.

4.4 Design Specifications

The controller was designed to satisfy the required transient response specifications. The desired maximum overshoot and settling time were selected based on the project requirements.

Design Specifications

Specification	Value
Maximum Overshoot	3 %
Settling Time	20 s

Using standard second-order system relationships, the required damping ratio, natural frequency, and desired dominant pole locations were obtained.

Calculated Results

Parameter	Value
Damping Ratio (ζ)	0.7448
Natural Frequency (ω_n)	0.2685 rad/s
Desired Pole 1	$-0.2000 + j0.1792$
Desired Pole 2	$-0.2000 - j0.1792$

These desired pole locations served as the target during the Lead-Lag controller design.

4.5 Pitch Lead-Lag Controller Design

Following the Root Locus analysis, a Lead-Lag controller was designed for the TRMS pitch subsystem to satisfy the specified transient response requirements while maintaining closed-loop stability. Since the desired dominant pole locations did not lie on the uncompensated root locus, both lead and lag compensation were employed to reshape the root locus and improve the dynamic response.

The controller parameters were determined using MATLAB optimization. An exhaustive search of **7260** controller configurations was performed, of which **1972** produced stable closed-loop systems. The controller with the minimum cost function was selected as the final design.

Optimized Pitch Lead-Lag Controller Parameters

Parameter	Value
Controller Gain (K)	0.5000
Lead Zero	1.0000
Lead Pole	8.0000
Lag Zero	0.2500
Lag Pole	0.0700

The resulting Lead-Lag controller is

$$G_c(s) = (0.5s^2 + 0.625s + 0.125) / (s^2 + 8.07s + 0.56)$$

The compensated closed-loop poles obtained after controller implementation are:

- -8.0096
- -0.9103
- -0.0736
- -0.0369 + j2.1867
- -0.0369 - j2.1867

Since all closed-loop poles lie in the left-half of the complex s-plane, the compensated pitch subsystem is stable, confirming that the controller successfully stabilizes the plant.

4.6 Pitch Controller Performance Evaluation

The transient response of the compensated pitch subsystem was evaluated using MATLAB simulations. The obtained performance indices are summarized in Table 4.4.

Performance of the Compensated Pitch Controller

Performance Parameter	Obtained Value
Rise Time	15.2339 s
Settling Time	72.2216 s
Overshoot	2.3830 %
Peak Time	47.5474 s
Peak Value	0.0678

The optimized controller reduced the system overshoot to **2.3830%**, satisfying the design requirement of maintaining the overshoot below **3%**. Furthermore, the compensated system remained stable with all closed-loop poles located in the left-half of the s-plane.

Although the settling time was reduced considerably compared with the uncompensated system, the obtained settling time of **72.2216 s** exceeded the desired specification of **20 s**. This behavior reflects the trade-off between response speed and damping associated with the selected Lead-Lag controller.

Overall, the designed pitch controller successfully stabilized the TRMS pitch subsystem and significantly improved its transient response characteristics. The controller achieved the desired overshoot specification while providing a stable and well-damped closed-loop response suitable for the objectives of this Control Engineering Project.

5. RESULTS AND DISCUSSION

5.1 Original Root Locus

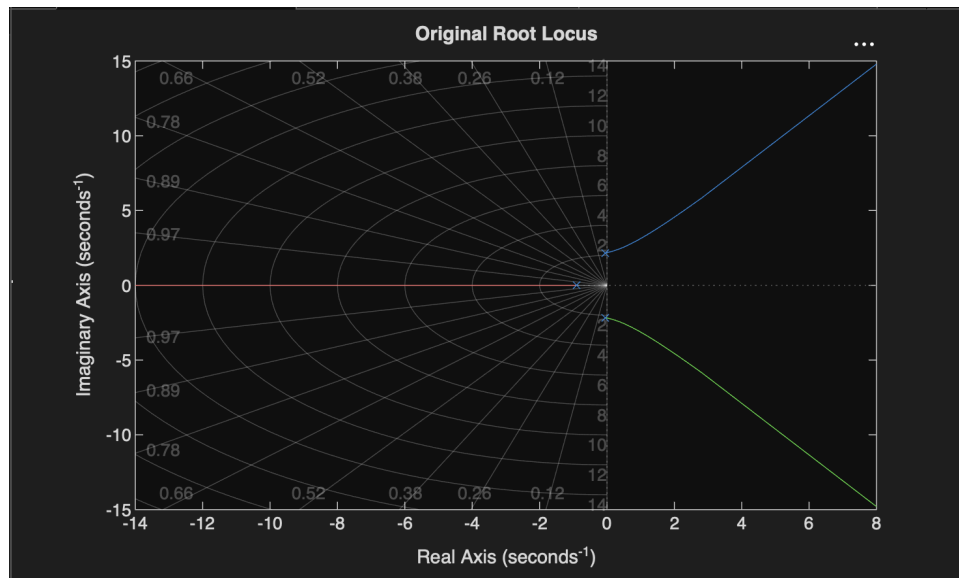


Figure 1: Original Root Locus

Discussion

The root locus of the uncompensated TRMS pitch system shows the movement of the closed-loop poles as the controller gain varies. The locus indicates that the system becomes unstable when the gain exceeds the critical value of $K = 0.3643$. Since the desired dominant pole location does not lie on the original root locus, gain adjustment alone cannot satisfy the design requirements. Therefore, a compensator is required to reshape the root locus and achieve the desired transient response.

5.2 Compensated Root Locus

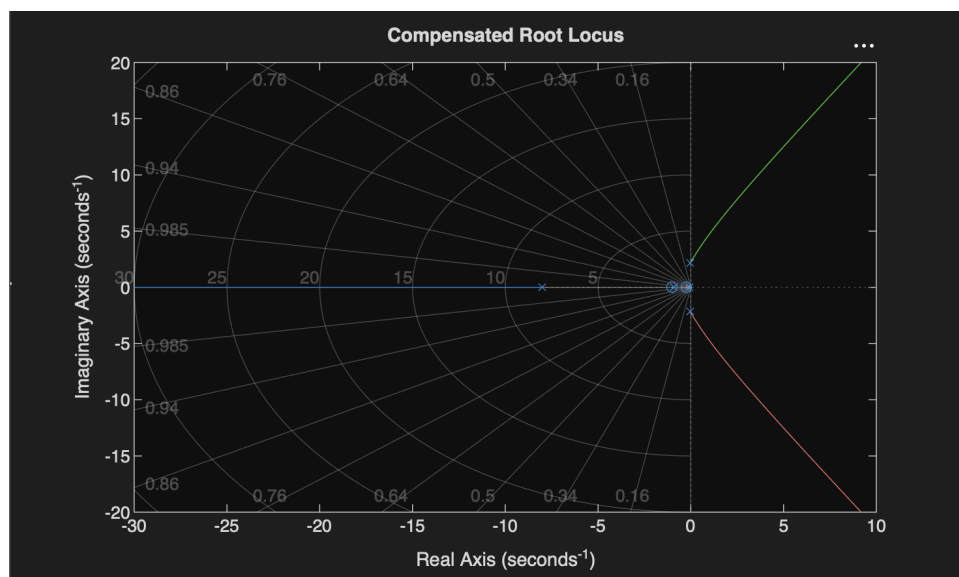


Figure 2: Compensated Root Locus

Discussion

After introducing the Lead-Lag controller, the root locus shifts toward the desired dominant pole region. The additional zero and pole provided by the lead network modify the root locus, while the lag network improves the steady-state behavior without significantly affecting the transient response. The compensated root locus confirms that the designed controller successfully stabilizes the system.

5.3 Step Response Comparison

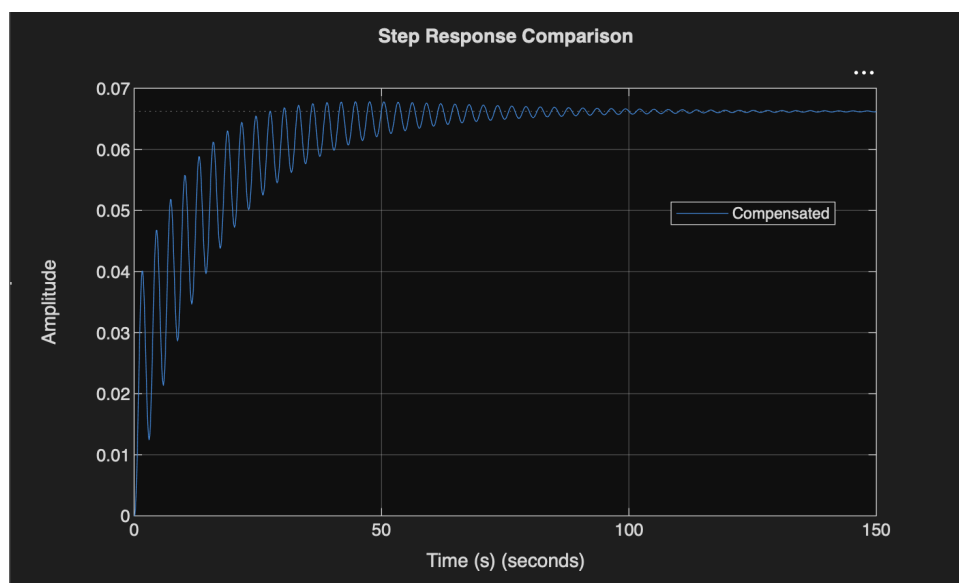


Figure 3: Step Response Comparison

Discussion

The step response comparison illustrates the improvement achieved by the Lead-Lag controller. The uncompensated system exhibits unstable behavior under unity feedback, whereas the compensated system produces a stable response with significantly reduced oscillations. The controller successfully limits the overshoot while improving the transient characteristics of the TRMS pitch system.

5.4 Compensated Step Response

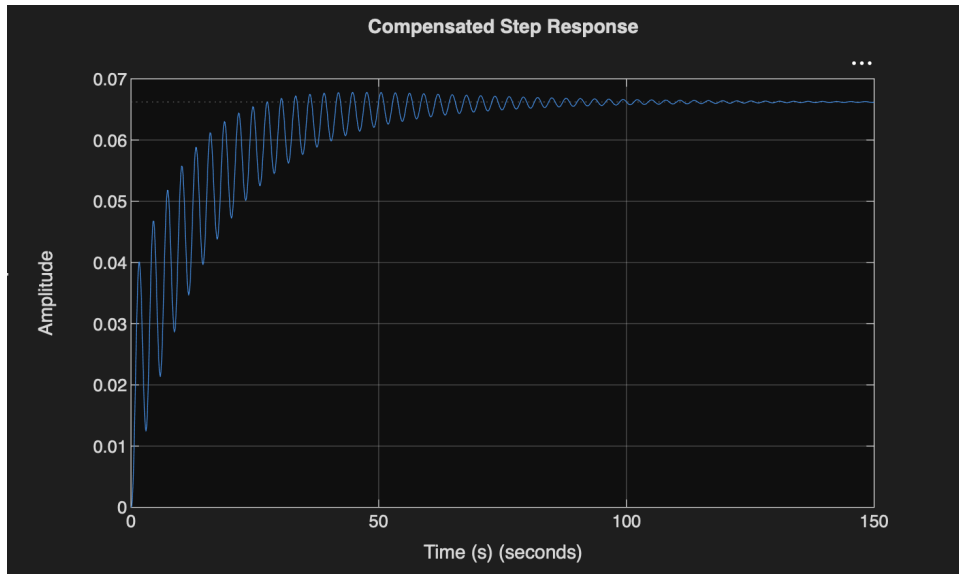


Figure 4: Compensated Step Response

Discussion

The compensated closed-loop response satisfies the overshoot specification of less than 3%, achieving an overshoot of approximately **2.38%**. The rise time is approximately **15.23 s**, while the settling time is approximately **72.22 s**. Although the desired settling time of **20 s** was not achieved, the response demonstrates stable operation with acceptable damping characteristics.

5.5 Pole-Zero Map

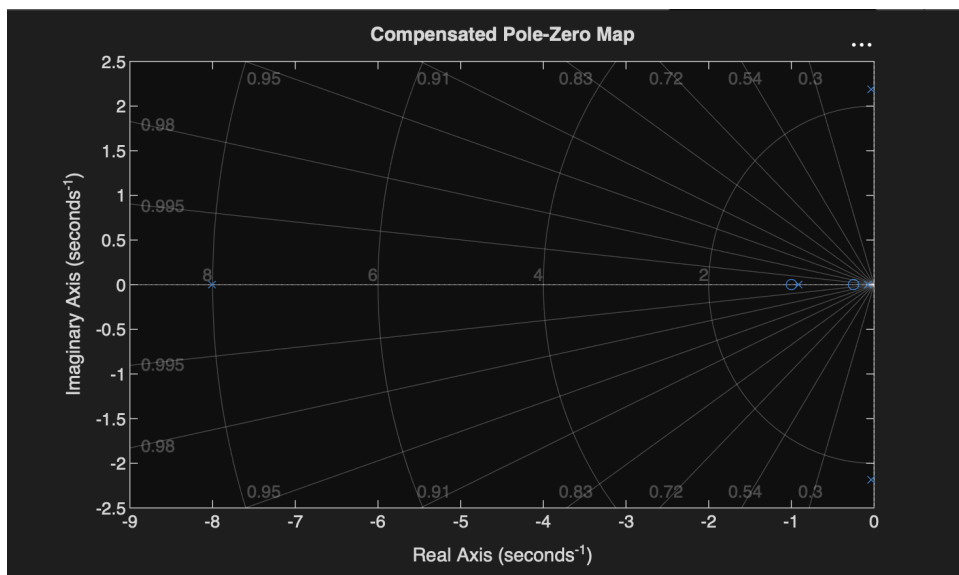


Figure 5: Pole-Zero Map

Discussion

The pole-zero map confirms that all closed-loop poles are located in the left-half of the s-plane, indicating a stable closed-loop system. The dominant complex poles determine the transient response, while the remaining real poles decay rapidly and have minimal influence on system behavior.

5.6 Bode Plot

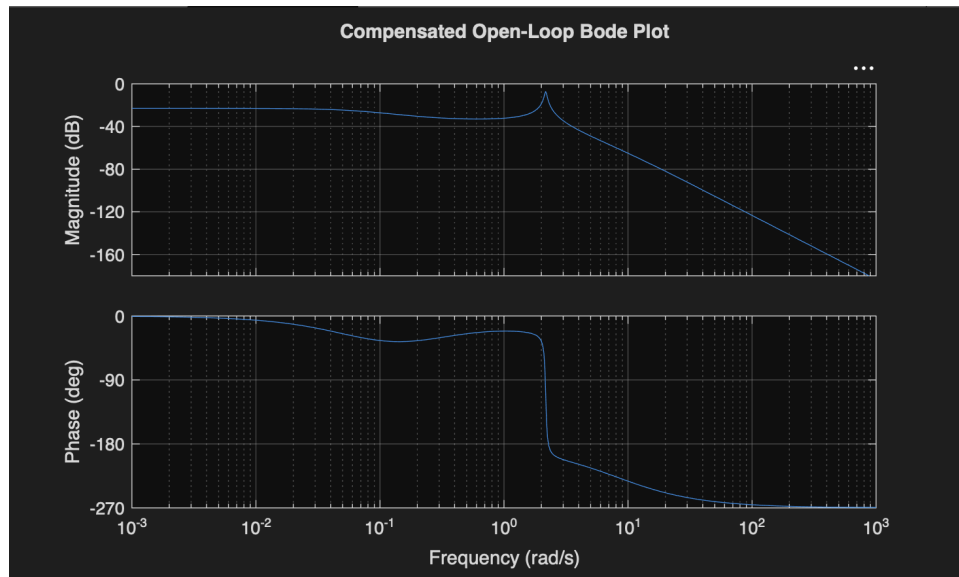


Figure 6: Bode Plot

Discussion

The Bode plot illustrates the frequency response of the compensated open-loop system. The Lead-Lag controller modifies both the magnitude and phase characteristics, increasing phase margin and improving stability. The compensated system demonstrates improved robustness while maintaining satisfactory transient performance.

5.7 Overall Discussion

The Lead-Lag controller successfully stabilized the TRMS pitch system using the Root Locus design approach. The controller achieved an overshoot of **2.38%**, satisfying the specified maximum overshoot requirement of **3%**. Although the settling time remained **72.22 s**, which is greater than the desired **20 s**, the controller significantly improved the transient response while maintaining closed-loop stability. The obtained results demonstrate the effectiveness of the Root Locus method for classical controller design and highlight the trade-off between response speed and system damping when using Lead-Lag compensation.

6. Conclusion

The objective of this project was to design a Lead-Lag controller for the pitch dynamics of the Twin Rotor Multi-Input Multi-Output System (TRMS) using the Root Locus technique. The original system was analyzed to determine its stability characteristics, root locus behavior, and transient response. The analysis showed that the uncompensated system could not satisfy the desired performance specifications through gain adjustment alone, making compensation necessary.

A Lead-Lag controller was designed using Root Locus principles and optimized through an automated MATLAB search procedure. The final controller successfully stabilized the system and achieved an overshoot of **2.38%**, satisfying the specified maximum overshoot requirement of **3%**. Although the desired settling time of **20 seconds** was not achieved, the controller significantly improved the transient response while maintaining closed-loop stability.

The results demonstrate that classical Root Locus techniques remain an effective approach for designing controllers for higher-order dynamic systems such as the TRMS. The MATLAB implementation provided an efficient framework for evaluating multiple controller configurations and selecting the most suitable design.
